

JOBMATCH AI

Official Development Work Plan

Version 1.0

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1. Project Overview

JobMatch AI is an AI-powered employment and skills matching platform designed to automate:

- CV parsing and skill extraction
- Job–candidate matching
- Skills gap analysis
- Training recommendations
- Career guidance chatbot
- Web and mobile user interfaces

The system is divided into five core modules:

1. Backend
2. AI Engine
3. Frontend
4. UI/UX
5. Mobile App

Each module has a dedicated team and folder structure.

2. Project Folder Structure

```
JobMatch_AI/  
├── AI_Engine/  
├── Backend/  
├── Frontend/  
├── UI_UX/  
└── Mobile_App/
```

3. AI Engine Module

Team: AI Developers

Priority: High (after backend)

3.1 Purpose

The AI Engine provides all intelligence for JobMatch AI:

- Resume parsing
- Skill extraction
- Job matching
- Skills gap detection
- Training recommendations
- Career chatbot
- Embedding models

3.2 Folder Structure

```

AI_Engine/
├── resume_parser/
│   ├── cv_extractor.py
│   ├── skill_detector.py
│   └── education_extractor.py
├── matching_engine/
│   ├── job_matcher.py
│   ├── embeddings.py
│   └── similarity_model.py
├── skill_gap_analysis/
│   ├── gap_detector.py
│   └── skill_database.py
├── recommendation_system/
│   ├── course_recommender.py
│   └── training_mapping.py
├── chatbot/
│   ├── career_bot.py
│   └── conversation_model.py
├── models/
│   └── saved_models/
├── datasets/
├── requirements.txt
└── README.md

```

3.3 AI Developer Responsibilities

AI Developer 1 — Resume Parser

Deliverables:

- CV text extraction
- Skill extraction
- Education extraction
- Experience extraction

Output Format (JSON):

```
{ "name": "Yahya Sarr", "skills": ["Python", "Database", "Machine Learning"]
}
```

AI Developer 2 — Job Matching Engine

Deliverables:

- Embedding model

- Similarity scoring
- Match percentage

Example Output: Match Score: 82%

AI Developer 3 — Skills Gap + Training Recommendation

Deliverables:

- Missing skills detection
- Course recommendation engine

AI Developer 4 — Career Chatbot

Deliverables:

- Conversational model
- Career guidance responses
- Learning roadmap generator

4. Backend Module

Team: Backend Developers

Priority: Highest

4.1 Purpose

The backend is the core system responsible for:

- Authentication
- User management
- Job management
- Applications
- Employer tools
- AI Engine communication
- Database (PostgreSQL)
- REST APIs

4.2 Folder Structure

Backend/

```

|
├── authentication/
│   ├── login
│   ├── register
│   ├── password_reset
│   └── email_verification
├── users/
│   ├── models.py
│   └── views.py
└──

```

```

|
|— jobs/
|   |— job_models.py
|   |— job_views.py
|   |— job_search.py
|
|— applications/
|   |— apply.py
|   |— tracking.py
|
|— employers/
|
|— ai_connection/
|   |— ai_api.py
|
|— notifications/
|
|— database/
|
|— settings/
|
|— requirements.txt

```

4.3 Backend Responsibilities

User Management

- Job seeker registration
- CV upload
- Profile management
- Employer accounts
- Admin controls

Job System

Employer:

- Create job
- Edit job
- Delete job
- View applicants

Candidate:

- Search jobs
- Apply
- Track application

API Endpoints

- POST /register
- POST /login
- GET /jobs

- POST /apply
- GET /match-score

5. Frontend Module

Team: Web Developers

Technology: React

5.1 Folder Structure

```
Frontend/  
├── src/  
├── components/  
├── pages/  
├── authentication/  
├── dashboard/  
├── jobs/  
├── profile/  
├── chatbot/  
├── services/  
└── assets/
```

5.2 Required Pages

Landing Page

- Project overview
- Login/Register

Job Seeker Dashboard

- Profile
- CV upload
- Recommended jobs
- Match percentage
- Training recommendations

Employer Dashboard

- Post jobs
- View candidates
- Applicant ranking

Admin Dashboard

- User management
- Reports
- Analytics

6. UI/UX Module

Team: Designers

Tools: Figma, Adobe XD

6.1 Folder Structure

```
UI_UX/  
├── wireframes/  
├── prototypes/  
├── user_flows/  
├── design_system/  
├── icons/  
├── screenshots/  
└── documentation/
```

6.2 Deliverables

- Wireframes
- High fidelity prototypes
- User flows
- Design system
- Screens for all pages

7. Mobile App Module

Team: Mobile Developers

Technology: Flutter

Priority: Last (after web version)

7.1 Folder Structure

```
Mobile_App/  
├── lib/  
├── screens/  
├── widgets/  
├── authentication/  
├── api/  
├── offline_storage/  
└── assets/
```

7.2 Features

Job Seeker

- Register
- Upload CV
- Search jobs

- Apply
- Chat
- View recommendations

Employer

- Post jobs
- Review candidates
- Chat

8. Development Priority Order

Priority	Module	Reason
1	Backend	All systems depend on APIs
2	AI Engine	Core innovation
3	Frontend	User-facing system
4	UI/UX	Supports frontend
5	Mobile App	Build after web version

9. Team Assignments

Team	Responsibility
AI Team	CV parsing, matching, chatbot
Backend Team	APIs, database, authentication
Frontend Team	React web app
UI/UX Team	Design system, prototypes
Mobile Team	Flutter app

10. Exhibition Demo Plan

To impress judges:

Demo Flow

1. Upload CV
2. AI extracts skills
3. AI generates match score
4. Recommended jobs appear instantly

